Material Data Sheet, February

TECAFLON PTFE

Chemical Designation :	Polytetrafluorethylene
DIN−Abbreviation:	PTFE
Colours, fillers:	opaque

Main features

- excellent chemical resistance
- excellent sliding properties
- inherently flame retardant (UL94 V−0)
- very good electrical insulation
- very good UV−resistance
- continuous service temperature up to 260°C

Preferred Fields

- chemical engineering
- transport and conveyor technology
- Semiconductor technology
- cryogenics
- medical technology
- mechanical engineering
- pumps and instrument manufacture
- pure water production
- food technology

Applications

pump housings, filter housings, valve seats, etching plates, tank linings, high frequency insulators, pipe linings, pump rotors, roller coverings, seals, plain bearings (housed) slide runners

Properties

Mechanical	dry / moist		standard
Tensile strength at yield	25	MPa	DIN EN ISO 527
Elongation at yield		%	

Tensile strength at break		MPa	
Elongation at break	> 50	%	DIN EN ISO 527
Modulus of elasticity in tension	700	MPa	DIN EN ISO 527
Modulus of elasticity after flexural test		MPa	
Hardness	30		DIN 53 456 (Kugeldruckhärte)
Impact strength 23º C (Charpy)	n.b.	KJ/m²	DIN EN ISO 179 (Charpy)
Creep rupture strength after 1000 h with static load	5	MPa	
Time yield limit for 1% elongation after 1000 h	1,58	MPa	
Co−efficient of friction p = 0,05 N/mm²v=0,6 m/s on steel, hardened and ground	0,08−0,10		
Wear p = 0,05 N/mm²v=0,6 m/s on steel, hardened and ground	21	µm/km	

Thermal	**dry / moist**		**standard**
Crystalline melting point		°C	
Glass transition temperature	−20	°C	DIN 53 765
Heat distortion temperature HDT, Method A	55	°C	ISO−R 75 Verfahren A (DIN 53 461)
Heat distortion temperature HDT, Method B	121	°C	ISO−R 75 Verfahren B (DIN 53 461)
Max. service temperature			
short term	260	°C	
long term	260	°C	
Thermal conductivity (23º C)	0,25	W/(K·m)	
Specific heat (23º C)	1	J/g.K	
Coefficient of thermal expansion (23−55ºC)	12	10^{-5} 1/K	DIN 53 752

Properties

Electrical	dry / moist		standard
Dielectric constant (10^6 Hz)	2,1		DIN 53 483, IEC−250
Dielectric loss factor (10^6 Hz)	0,0002		DIN 53 483, IEC−250
Specific volume resistance	10^16	Ω*cm	DIN IEC 60093
Surface resistance	10^16	Ω	DIN IEC 60093
Dielectric strength	48	kV/mm	DIN 53 481, IEC−243, VDE 0303 Teil 2
Resistance to tracking	KA 3c KB>600		DIN 53 480, VDE 0303 Teil 1

Miscellaneous	dry / moist		standard
Density	2,18	g/cm^3	DIN 53 479
Moisture absorption (23°C/50RH)	< 0,05	%	DIN EN ISO 62
Water absorption to equilibrium		%	DIN 53 495
Flammability acc. to UL standard 94	V0		

(1) Testing of semi−finished products

The above information corresponds with our current knowledge and indicates our products and possible applications. We cannot give a legally binding guarantee of chemical resistance, of certain properties and the suitability of our products and their applications. Our products are not destined for use in medical and dental implants. Existing commercial patents must be observed. Unless otherwise stated, these values represent averages taken from injection moulding samples, dry as moulded. We reserve the right to make technical alterations.